

It's a Small World After All¹

Your name:

Who you are working with:



Warm-Up

First let's review some stuff that we did yesterday and make things a bit more general:

1. What is $10^8 \cdot 10^{12}$ as an exponent? (We want to avoid writing lots of zeros!)
2. Is it true that $10^a \cdot 10^b$ equals $10^b \cdot 10^a$? Why or why not? Can you support your answer with an example? (Let a and b just be whole numbers for now. Let's see some words for this answer!)
3. What is $10^a \cdot 10^b \cdot 10^c$ equal to as an exponent?
4. What is $((10^2)^3)^4$ written as an exponent (without brackets)?
5. Is it true that $(10^a)^b$ equals $(10^b)^a$? Why or why not? Can you support your answer with an example? (Let a and b just be whole numbers for now. Let's see some words for this answer!)
6. What is $((10^a)^b)^c$ equal to as an exponent without brackets?
7. What is $(10^p \cdot 10^q)^r$ equal to as an exponent without brackets?

¹Part 2 of An introduction to Scientific Notation and Scale in five parts, written by Joshua Aaron, Comic by Bill Waterson.

Sticking up for the Little Guy

Now we are going to get into some deep philosophical stuff—we have been using exponents that are only whole numbers at this point, but it turns out this whole “put a little number in the upper right hand corner” notation is *way* more powerful than that! To start, let’s consider the following:

$$\frac{10}{10} = \frac{1}{1} = 1 \quad \frac{10^2}{10^2} = \frac{1}{1} = 1 \quad \frac{10^n}{10^n} = \frac{1}{1} = 1$$

But we also know from yesterday and our warm-up that the following is true:

$$10^a \cdot 10^b = 10^{a+b}$$

Key Question! What should I put in instead of the question mark (“?”) for the following equation to be true?

$$10^a \cdot 10^7 = 10^0 = 1$$

To help with this, let’s work on the following equations:

1. Solve for x in the equation $10^3 \cdot 10^x = 10^0 = 1$: (Check out the rules we made in terms of a and b earlier! :))
2. What are the following equal to? $\frac{10^7}{10^4}$ and $\frac{10^4}{10^7}$.
3. What is $\frac{10^7}{10^4} \cdot \frac{10^4}{10^7}$ equal to?

DANGER! DO NOT READ PAST THIS POINT UNTIL YOU FINISH THE ABOVE QUESTIONS! YOU COULD POTENTIALLY CAUSE A RIP IN THE SPACE-TIME CONTINUUM AND DESTROY THE UNIVERSE!

So now that you have done the above problems you have hopefully generated the idea of idea of *negative exponents*! So what do we do with this? Well, you might know that 1 centimeter is one $\frac{1}{100}$ th of a meter, and that a millimeter is one $\frac{1}{1000}$ th of a meter, but now we can write these another way:

$$1cm = \frac{1}{100}m = \frac{1}{10^2}m = 10^{-2}m \quad \text{and} \quad 1mm = \frac{1}{1000}m = \frac{1}{10^3}m = 10^{-3}m$$

Tomorrow we will go through even more prefixes that help us measure things at larger and smaller scales, but for now I will leave you with three mind blowing questions:

1. A period at the end of a sentence (or the dot in a question mark, etc.) is about a tenth of a millimeter or $\frac{1}{10} \cdot 10^{-3}m = 10^{-4}m$. A *nanometer* (nm) is $10^{-9}m$ or one *billionth* of a meter! the smallest viruses are about 20nm ($20 \cdot 10^{-9}m$) in diameter—how many viruses can fit on the dot in the question mark at the end of this sentence?
2. One of the larger types of atoms we know about is Cesium (a super interesting element). It happens to be about 225 *picometers* in diameter, that is one *trillionth* (10^{-12}) of a meter (so we have $225 \cdot 10^{-12}m$). How many times bigger is the smallest virus compared to the larger atom? How many cesium atoms can fit in a period?
3. A tennis ball made of cesium atoms² would have a *radius* of (around) 6.75 Centimeters ($6.75 \cdot 10^{-2}m$). How many Cesium atoms make up this ball? (Remember, the diameter is twice the radius!)

²I do not recommend playing tennis with such a ball...